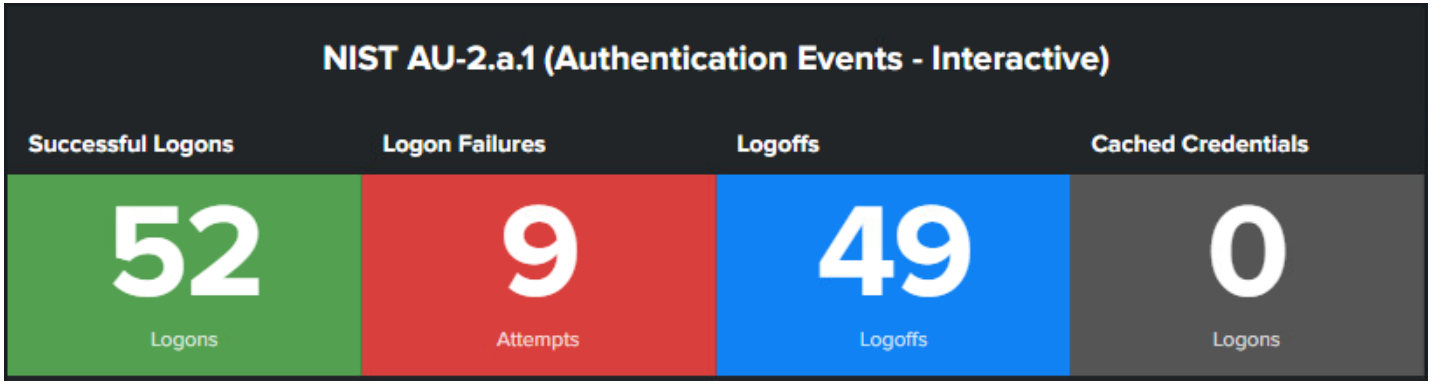
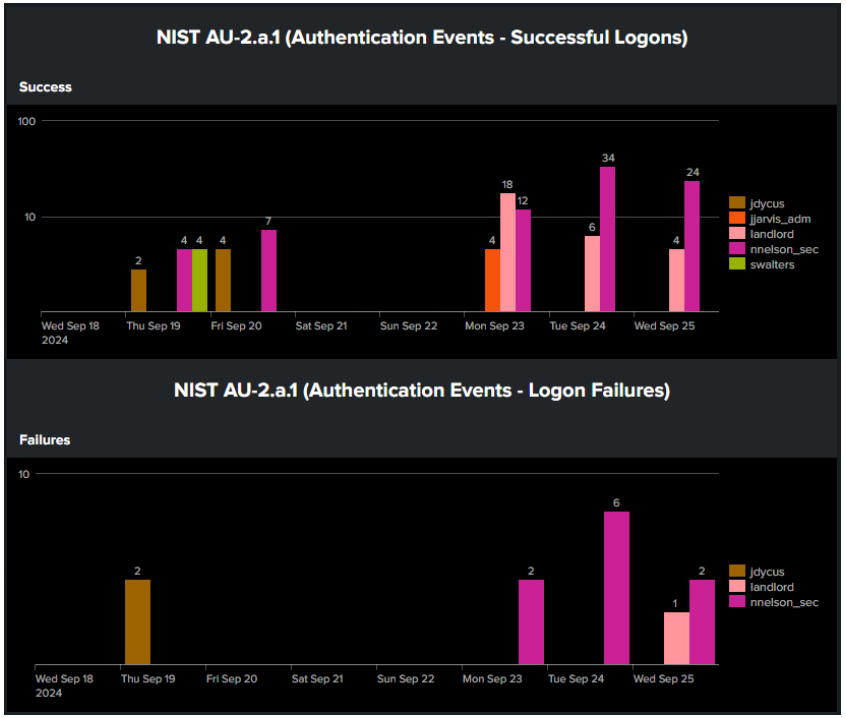
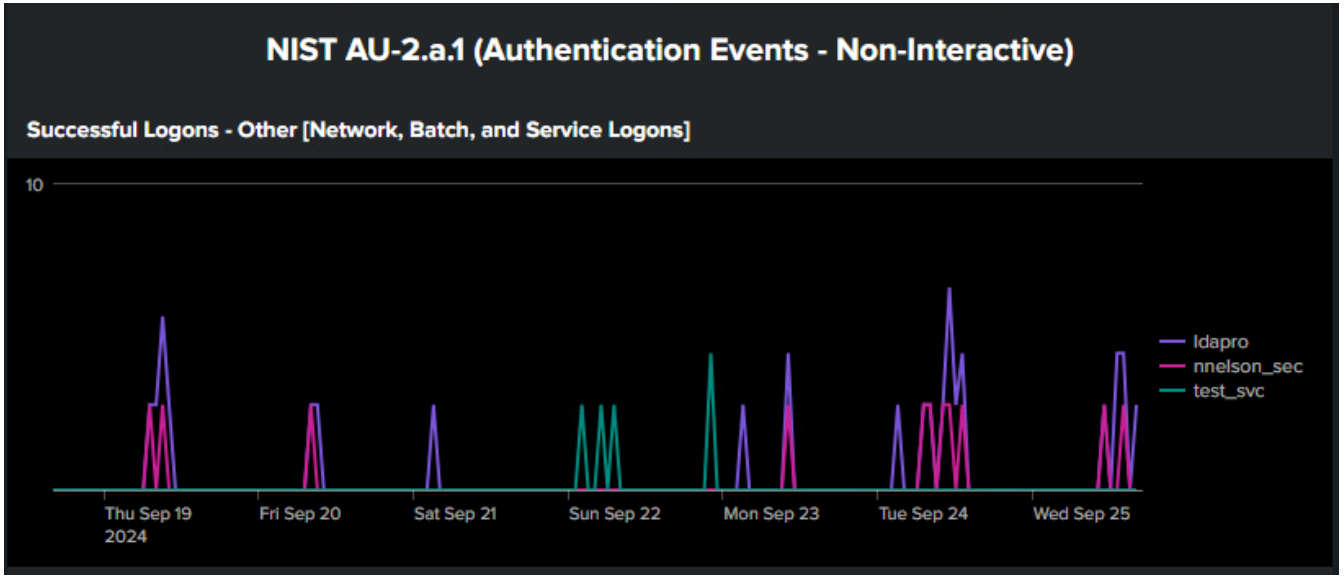
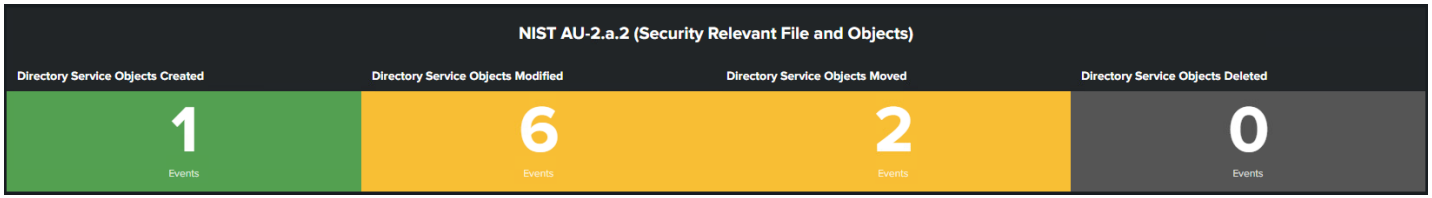
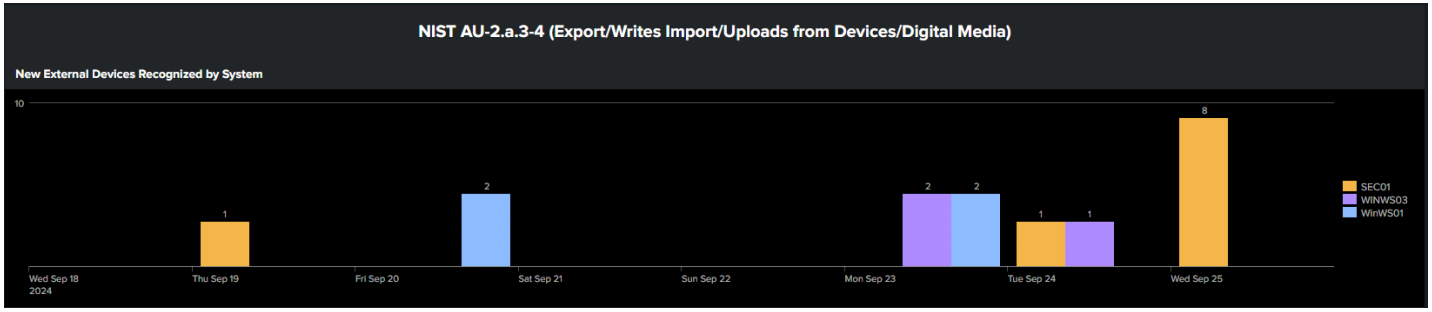
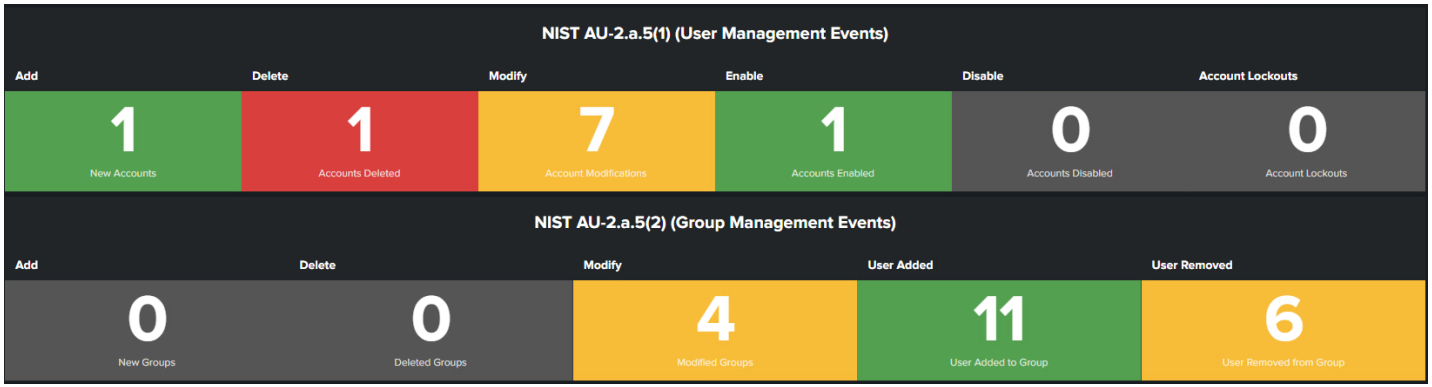
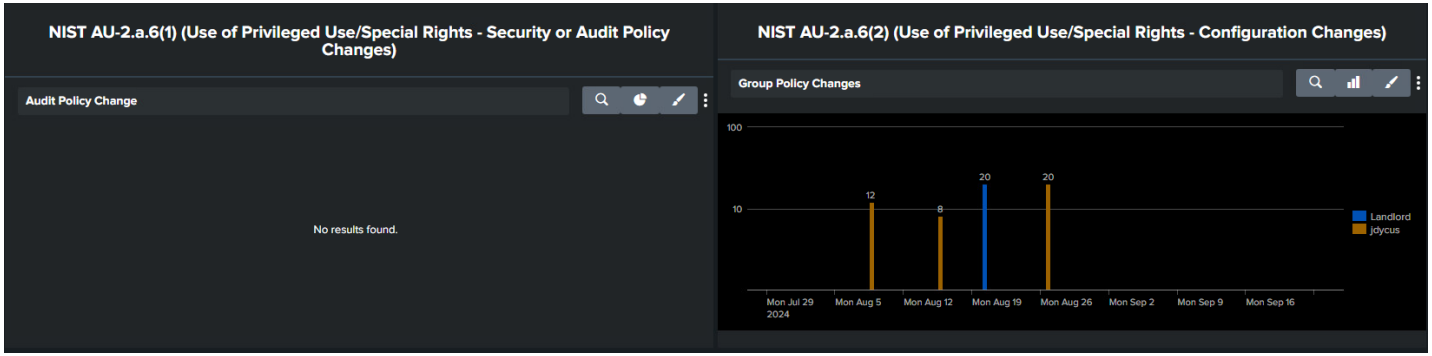
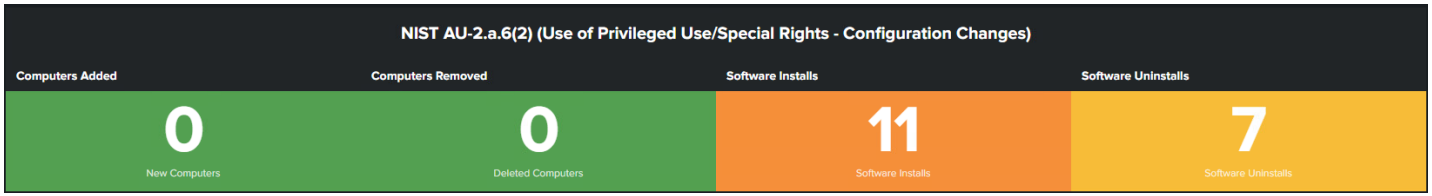
Cybersecurity App - Windows - Executive Summary

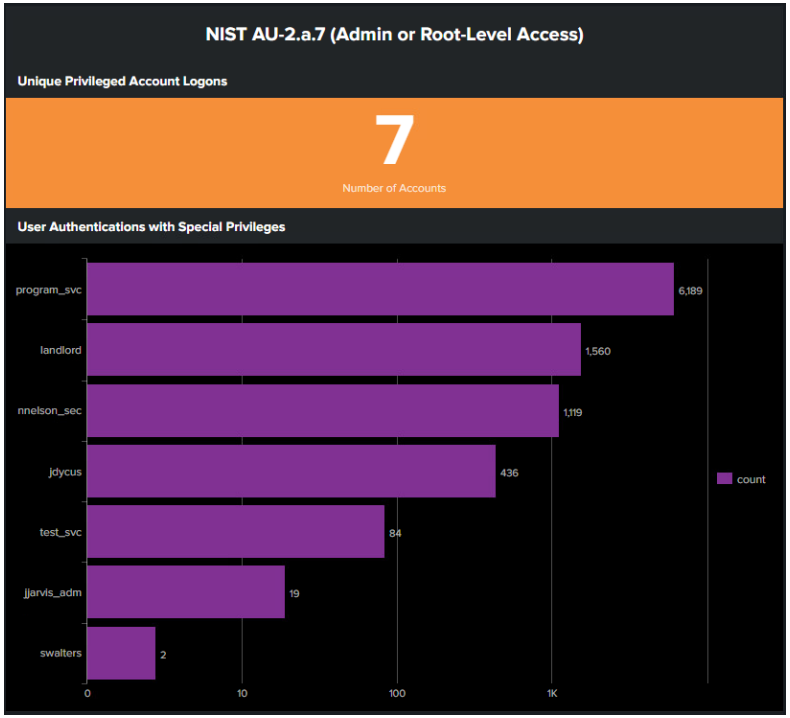
Panel	Windows - Executive Summary
NIST AU-2.a.1 (Authentication Events - Interactive)	This panel captures Windows-based interactive logins which includes: • Interactive (Console) Login • Unlocks • Remote Interactive Login The search also looks for Domain accounts, so system-level accounts will be filtered out. More on that in a later dashboard. Notable events: • Spikes could indicate brute force attempts • Low numbers could indicate there's a problem with forwarding logs • Cached Credentials (not including test equipment) should be investigated
	
NIST AU-2.a.1 (Authentication Events - Successful Logons) NIST AU-2.a.1 (Authentication Events - Logon Failures)	This panel displays interactive logins and attempts over the course of a 7 day period (See above panel). Notable events: • Inconsistent logons • Many failed attempts in one day • Weekend logons or attempts

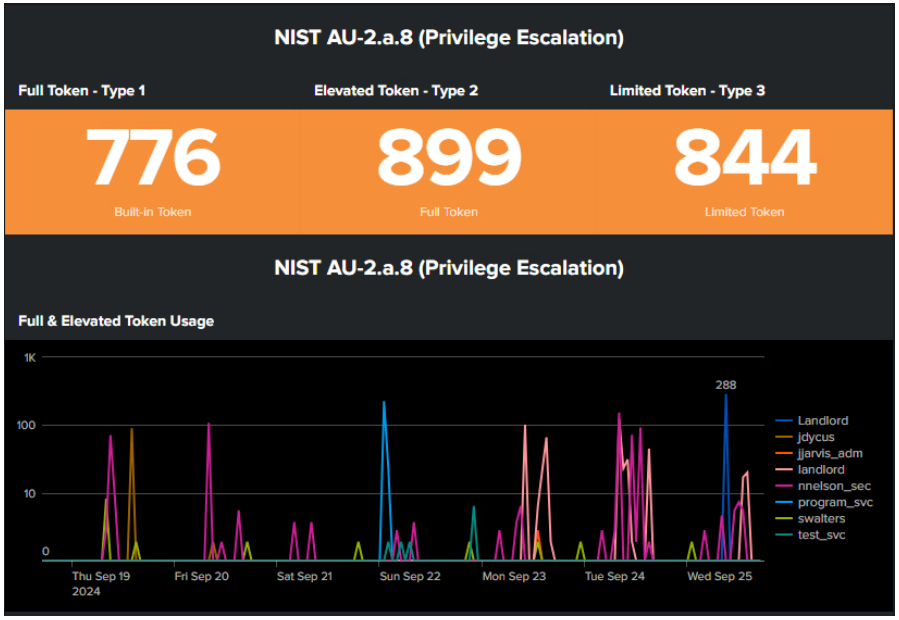
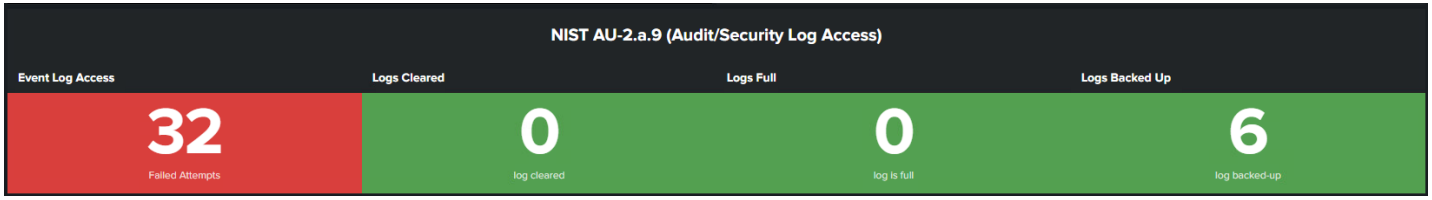
	
NIST AU-2.a.1 (Authentication Events - Non-Interactive)	This panel looks at network authentications and scripted or automated authentications. These will trigger as a response to an action, such as a scheduled task, accessing a network resource, or running scripts that require authentication. Notable events: • Spikes of activity • General user activity (high or consistent usage could be an anomaly) • Inconsistent Service Account authentications (consider patching over the weekend)

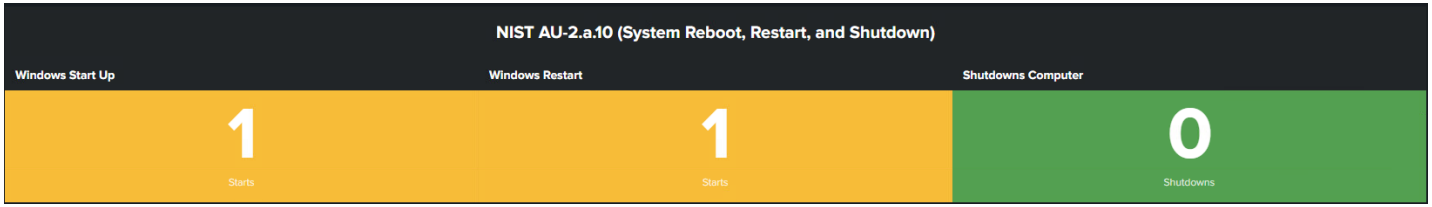
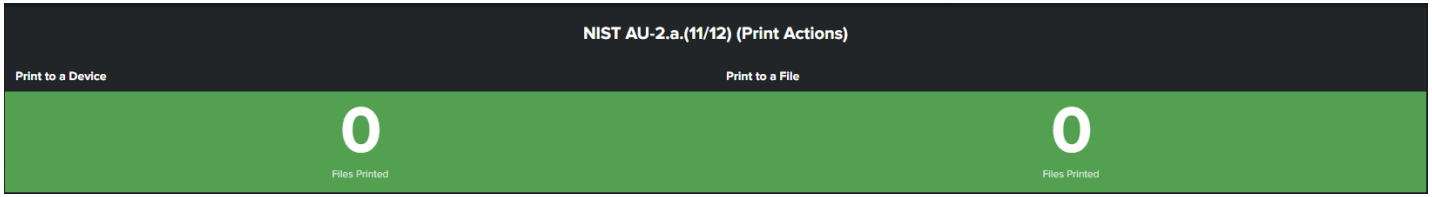
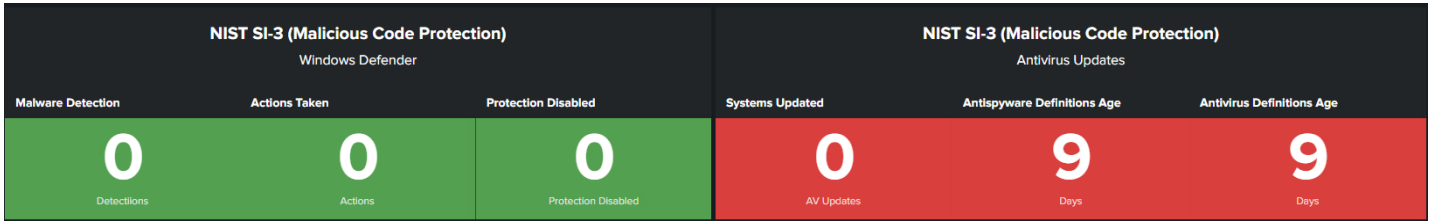
	
NIST AU-2.a.2 (Security Relevant File and Objects)	This panel shows events related to Security-Relevant Objects. This panel can be adjusted to include more objects that Security may be interested in. For example: Media Logs, Security/IT Folders, etc. Currently, only Active Directory Objects are listed, but later versions of Splunk will include more objects as requested. These events trigger when new accounts are created, computer objects are deleted, account settings are manually changed, etc. Notable events: • High numbers of objects created, modified, moved, or deleted • Look for users changing Active Directory objects without notice
	
NIST AU-2.a.3-4 (Export/Writes Import/Uploads from Devices/Digital Media)	This panel shows events that the system has logged when new devices are recognized (example: USB and removable media). This is only a system-level event that is logged, so user data is not associated with this event. Further drilling down into these device detections and login correlation can help track the user responsible for plugging in media. See Windows - Data Transfers for more granular audit events. Notable events: • Weekend activity • High number of devices plugged in • Devices recognized on servers • Devices plugged into multiple systems on the same day

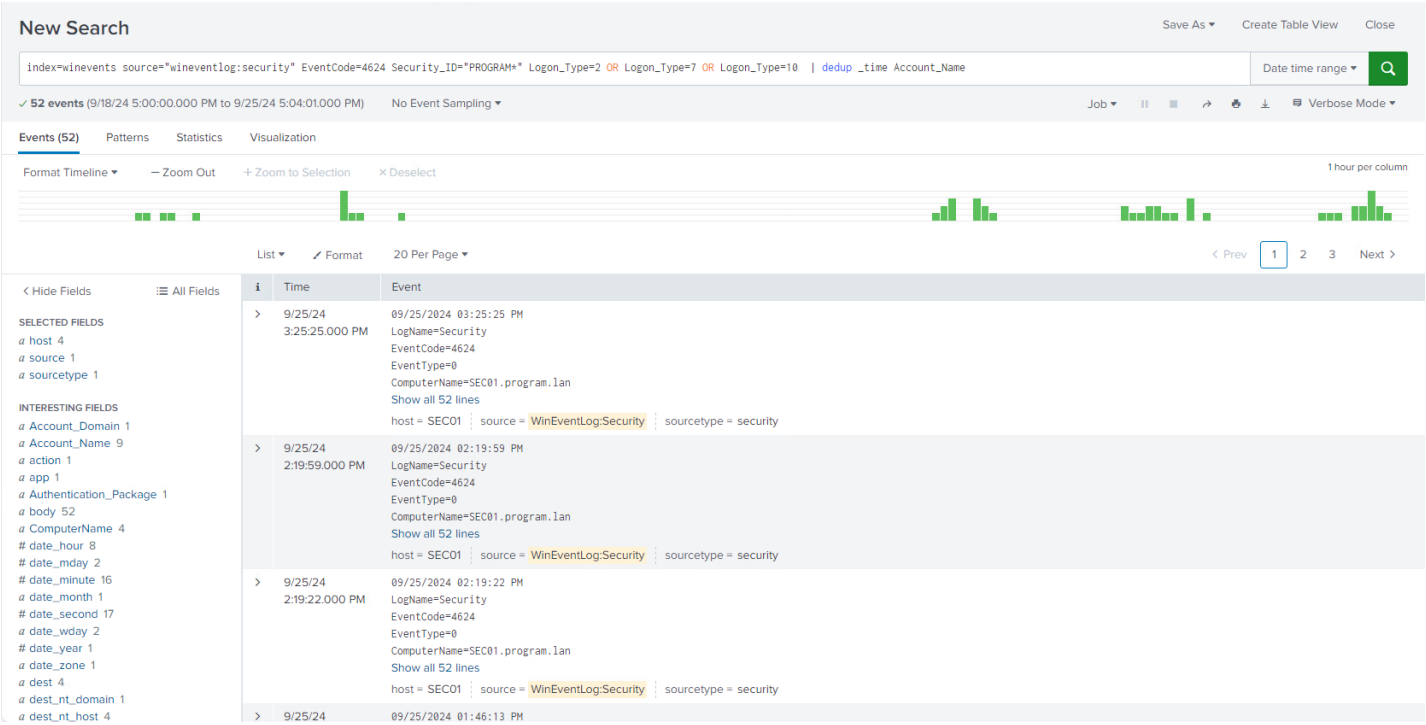
	
NIST AU-2.a.5(1) (User Management Events) NIST AU-2.a.5(2) (Group Management Events)	This panel shows account management activity (both local and domain-level accounts). System-level accounts generate very high levels of account management logs, so they will be filtered out of this panel. Most activity will be in Active Directory (for LANs and WANs). Notable events: • Accounts being deleted (typically they are only disabled) • High numbers of events Creation and deletion events should be more rare than modifications
	
NIST AU-2.a.6(1) (Use of Privileged Use/Special Rights - Security or Audit Policy Changes) NIST AU-2.a.6(2) (Use of Privileged Use/Special Rights - Configuration Changes)	This panel shows any changes to local security or audit policy and configuration changes. Since most of these events are handled via Group Policy, it should be rare that any events occur. Security and audit policy events are listed by the host name that the changes affect (pie chart visual). Notable events: • Any change in security or audit policy • High numbers of configuration changes • Non-IT personnel making configuration changes

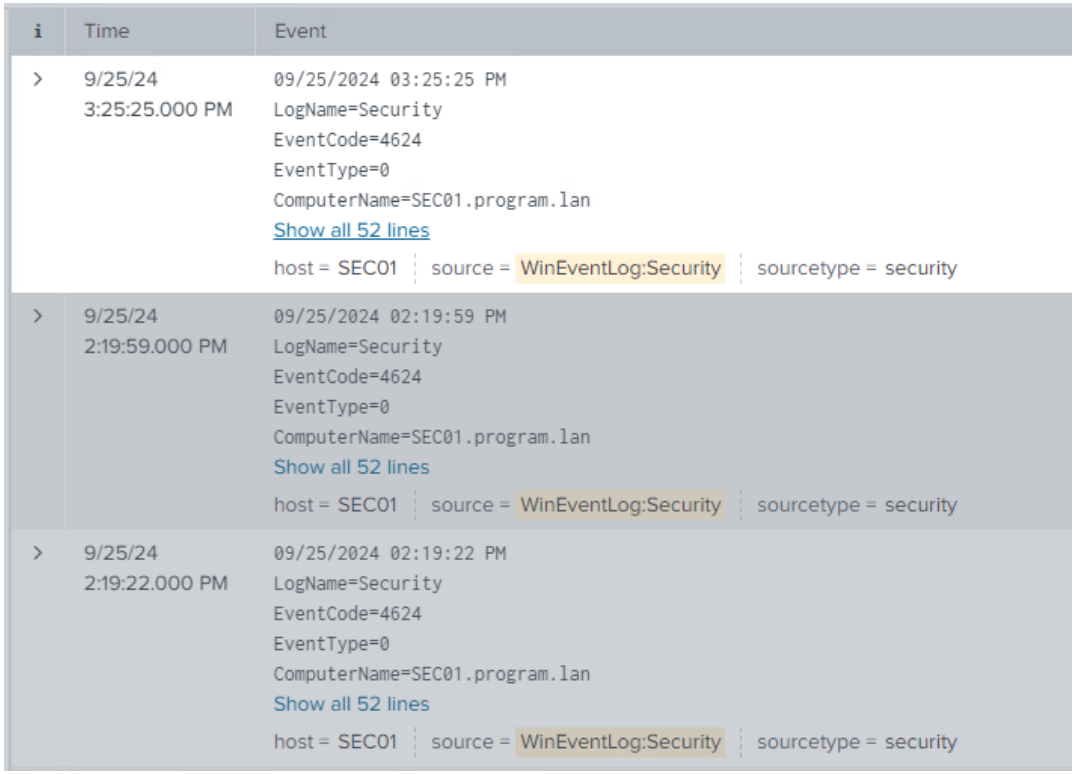
	
NIST AU-2.a.6(2) (Use of Privileged Use/Special Rights - Configuration Changes)	This panel goes into specific events relating to configuration changes. This panel can be adjusted to add more types of events in later versions of Splunk. Notable events: • High numbers of software installations or uninstalls • Any new or deleted computers (objects)
	
NIST AU-2.a.7 (Admin or Root-Level Access)	This panel displays accounts that have privileges assigned to them at login or via token manipulation. The first metric represents just the number of accounts with any amount of privileged events. The second summarizes the number of times those privileges were called by user account. Notable events: • Any user that shouldn't have admin privileges on a Windows system • High numbers of events associated with a specific user • High number of accounts with privileges assigned

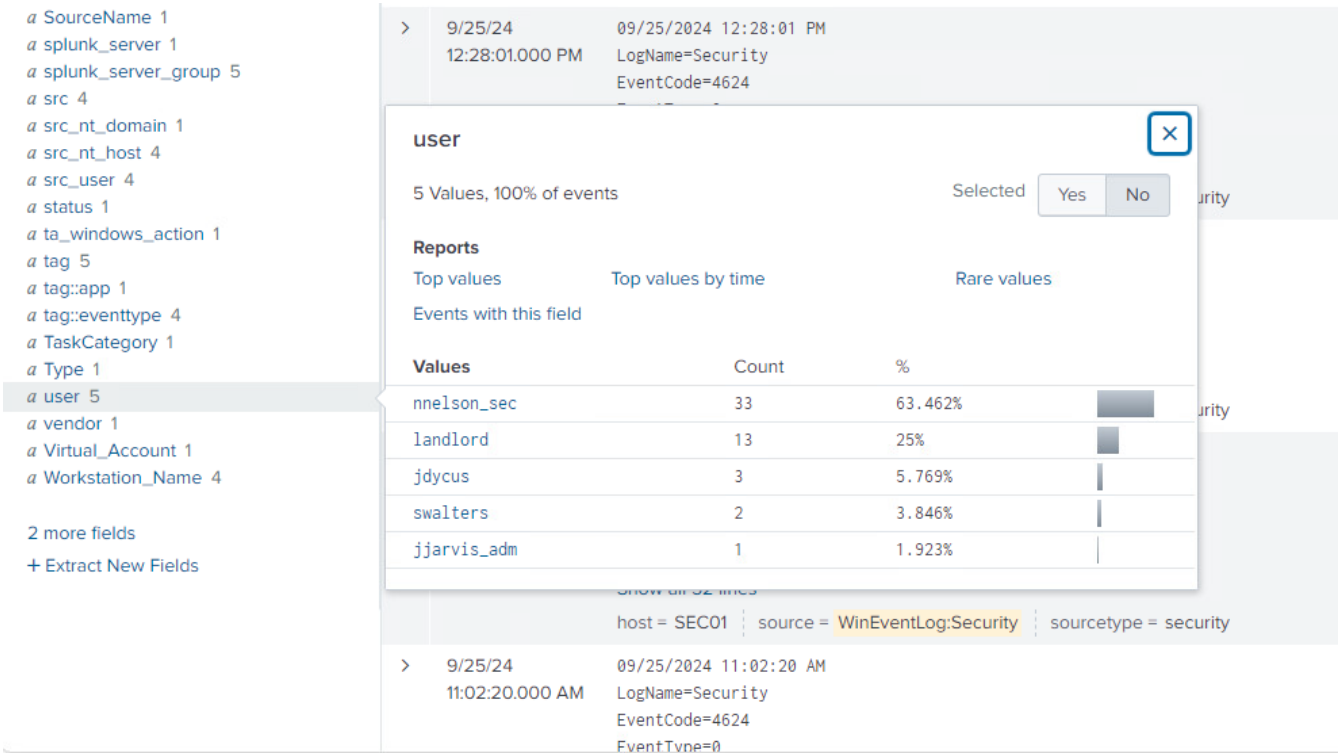
	
NIST AU-2.a.8 (Privilege Escalation)	This panel captures token usage by account and across the network. Tokens are how accounts are able to elevate privileges on Windows. Type 1 is considered the highest level token and the most privilege assigned, often bypassing UAC. Type 2 is the standard elevated token which most domain users will use. Type 3 tokens are for accounts that did not elevate (for that particular action or event that required authentication). Notable events: • Its rare that a user has Type 1 tokens • Spikes in token usage from unexplained accounts • Potential services running over the weekend in user's names or accounts that haven't logged out • Token manipulation by non-privileged accounts

	
NIST AU-2.a.9 (Audit/Security Log Access)	This panel shows events relating to Windows Event Logs. Logs should be periodically backed up, the other three events are notable concerns. Notable events: • High levels of failed attempts • Any logs that have been cleared • Logs full indicate that the system has stopped logging • High levels of logs backed up (might point to a misconfigured backup process)
	
NIST AU-2.a.10 (System Reboot, Restart, and Shutdown)	This panel shows system-level power events. Different types of shutdown events can occur depending on the circumstances (loss of power, user-initiated shutdown via the GUI, manual shutdowns via the power button). Notable events: • High levels of start ups or restarts (could indicate changes to the system) • Any shutdowns • Any events not done by Security or IT

	
NIST AU-2.a.11/12 (Print Actions)	This panel captures print jobs, both to a physical device and to a digital medium (PDF). Notable events: • Any prints to a physical device need to be accounted for and need control of that material • Very high levels of print to file might indicate malicious behavior
	
NIST SI-3 (Malicious Code Protection)	This panel displays basic information on Windows Defender, both scan results and definition updates. More information can be found on the Network Security dashboard. Notable events: • Malware Detected • AV Protection disabled on any system • No system updates within the last week • AV Definitions >7 Days old
	
Panel Drilldown	Description
Drilldown Search	When you select a result on any of the panels within this dashboard, you will be brought to a search for that panel. This is to offer flexibility and to be able to explore events natively. The time chart (green columns) allow you to hover over them to see times of events and to filter a search In this case if you select a column, you will see the events that occurred within the span of an hour **Any searches within this dashboard have pre-defined tables and searches under each target dashboard. You do not have to use this option for**

	most searches, but in the event of an investigation, you will need to know how to use the New Search page and filter for data to support the investigation.
	
Viewing Logs	Select the blue link Show all 52 lines to see the entire event log The > dropdown to the left will open the metadata that Splunk has tagged, including fields and values based on the format of the log

	 i | Time | Event 9/25/24 3:25:25.000 PM 09/25/2024 03:25:25 PM LogName=Security EventCode=4624 EventType=0 ComputerName=SEC01.program.lan Show all 52 lines host = SEC01 | source = WinEventLog:Security | sourcetype = security 9/25/24 2:19:59.000 PM 09/25/2024 02:19:59 PM LogName=Security EventCode=4624 EventType=0 ComputerName=SEC01.program.lan Show all 52 lines host = SEC01 | source = WinEventLog:Security | sourcetype = security 9/25/24 2:19:22.000 PM 09/25/2024 02:19:22 PM LogName=Security EventCode=4624 EventType=0 ComputerName=SEC01.program.lan Show all 52 lines host = SEC01 | source = WinEventLog:Security | sourcetype = security
Navigating fields	Selecting a field to the left (user) will display a pop up that identifies the top values for that field You can select a value (name) that is notable in your environment to filter for that specific field You can also select the Reports such as Rare values to see low occurrences of a particular event



Search Tips	• **Always specify an index** in your search, otherwise Splunk will have to search through billions of logged events (use index=winevents OR index=windows)
	  • I also like to specify the **source** or **sourcetype** to make the searches faster
	• Make sure the **time frame** isn't too large
	  • In most cases, within the last 90 days will be what I search for (especially for winevents)
	• The search bar allows you to do a **search for keywords**, in case the fields are not helping to identify information
	  • Wrap any terms you would like to search for in **"quotes"** and **make sure there are no pipes | before your search** or the query might fail
	  • Words that are in the middle of a string of text may need **wildcards** * on either side of the word "*Fail*"
	  • If you want to search more than one words, use **logical operators** between keywords (example: **AND, OR, NOT, IN**)